

Effects on Support Success in *Overwatch* 2

Experiment Background

My friends and I enjoy playing the hit game *Overwatch 2*. The main purpose of the game is to create a synergetic team consisting of high health/medium damage tanks, and medium health/high damage “dps” players, and low health/low damage support heroes that have the ability to heal their teammates. I have recently dedicated myself to learning a new support hero by the name of *Zenyatta*. *Overwatch 2* keeps track of in game statistics including damage done and healing done. Being able to maximize both of those as a support hero is key to in game success. After observing professional *Overwatch 2* players play as *Zenyatta*, I have narrowed down between 3 factors that vary the most between the professionals. Factor 1 is hero sensitivity. The low level represents a sensitivity level of 20% and the high is 80%. A low sensitivity will allow for better aim on targets and however, a higher sensitivity will allow for quicker movements, making the player less accurate but able to more quickly react to their environment. Factor 2 is positioning type. The low level is back line positioning. Back line positioning means that the player situates themselves close to their tank, which gives them protection from the enemy team. A flank play style on the other hand, sacrifices the safety of the tank on the hope that the player can catch the team at a spot they are not expecting, giving the player more chances to kill the enemy team. My final Factor is play style. Compared to most support heroes, *Zenyatta* offers more damage to opposing team than he does healing for his own. A heal play style will offer more support for fellow teammates at the expense of damage to the other team and vice versa.

Figure 1: Experiment Factors and Levels

Factor	Level	
	-	+
Sensitivity	20%	80%
Positioning Type	Back Line	Flank
Play Style	Damage	Heal

Experiment Setup

To test this experiment, I will use a full factorial design with 3 levels and 2 factors shown in Figure 1 resulting in 8 runs shown in Figure 2. To better normalize the data, each run will be conducted for a total of 5 times for a total trial that consists of 40 runs. The run order, also shown in Table 2, has a randomized run order in order to reduce learning effect for the one person running the 40 trials.

Figure 2: Experiment Planning Matrix

Run	Sensitivity	Positioning Type	Play Style	Run Order
1	20%	Back Line	Damage	7
2	20%	Back Line	Heal	3
3	20%	Flank	Damage	5
4	20%	Flank	Heal	1
5	80%	Back Line	Damage	6
6	80%	Back Line	Heal	4
7	80%	Flank	Damage	8
8	80%	Flank	Heal	2

Statistical Model

The means and variance represented by $\bar{y}$ and s^2 , where $\bar{y}$ is the mean for a run and s^2 is the variance for a run, represented by Equation 1 and 2, respectively.

Equation 1: Run Means

$$\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$$

Equation 2: Run Variance

$$s_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

To calculate the main effects of a factor, Equation 3 will be used. For the two-way interaction, Equation 4 will be used. All two-way interactions (AB AC AB) will be calculated using the same formula in Equation 4 but using the design matrix in figure 2. The three-way interaction of ABC will be calculated by using Equation 5.

Equation 3: Main Effects

$$ME(A) = \bar{y}(A+) - \bar{y}(A-) :$$

$$ME(B) = \bar{y}(B+) - \bar{y}(B-)$$

$$ME(C) = \bar{y}(C+) - \bar{y}(C-)$$

Equation 4: Two-Way Interaction Between Factor A and B

$$INT(A, B) = \frac{1}{2} [ME(A|B+) - ME(A|B-)]$$

Equation 5: Three-Way Interaction

$$INT(A, B, C) = \frac{1}{2} [INT(A, B|C+) - INT(A, B|C-)]$$

Data Analysis

After collecting all my data, I put my data into a .csv using excel and exported the .csv into r and read it using the built in read.csv() function.

Figure 3: Model Matrix of Zenyatta Experiment

Run	Sensitivity	Positioning Type	Play Style	Total Damage and Healing Done					$\bar{y}$	s^2	$\ln s^2$
1	-	-	-	6690	6422	7132	6221	7239	6740.8	193911.7	12.17515818
2	-	-	+	10704	12220	10471	11146	11749	11258	526338.5	13.17369982
3	-	+	-	5977	5734	6678	6305	6615	6261.8	164880.7	12.01297746
4	-	+	+	6570	6908	8134	7772	6723	7221.4	476735.8	13.07471774
5	+	-	-	9865	10245	10765	11043	11913	10766.2	618895.2	13.33569123
6	+	-	+	11682	12706	11895	12457	10817	11911.4	544918.3	13.20839115
7	+	+	-	5321	6420	7345	6742	7564	6678.4	785498.3	13.57407357
8	+	+	+	10436	9502	9302	9825	9585	9730	190913.5	12.15957572

The model matrix and factorial effects are represented by Figure 3 and Figure 4, respectively.

Figure 4: Factorial Effects for Zenyatta Experiment

Effect	$\bar{y}$	$\ln s^2$
A	1901	3.43
B	-2696.2	3.6
C	2418.4	3.55
AB	-438.4	2.69
AC	-320	2.54
BC	-412.8	2.66
ABC	1336	2.36

The factorial effects from Figure 4 show that factor B has the greatest impact on the total healing and damage done. Figure 3 and Figure 4 both show the ideal factor levels for maximizing damage and healing done is the level combinations of A+, B- and C+ which results in playing at a high sensitivity, in the back line, while focusing on healing. Figure 4 also shows us that the main effects of all A,B,C appear to be significant and the three-way effect of ABC may also be significant but is less clear.

Figure 5: Half-Normal Plot for Location

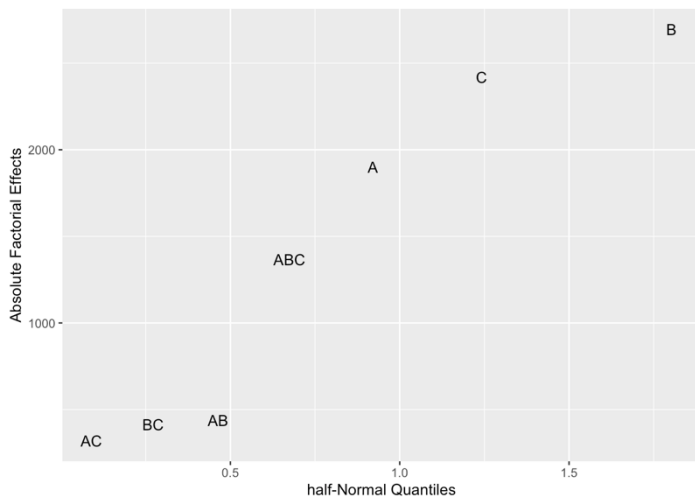
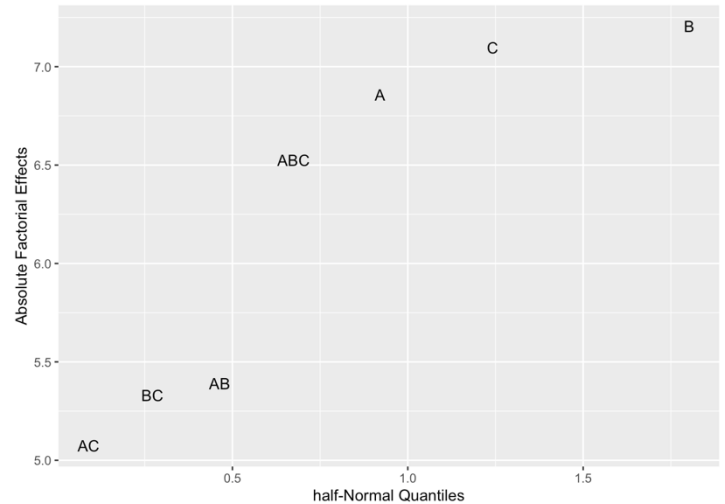


Figure 6: Half-Normal for Dispersion



Both of the half-normal plots for location and dispersion shown by Figure 5 and Figure 6, respectively, solidify the assertions made by the factorial effects. The main effects of A,B,C appear to be the most significant with the three-way factor of ABC also appearing to be significant.

Figure 7: Sensitivity vs Positioning

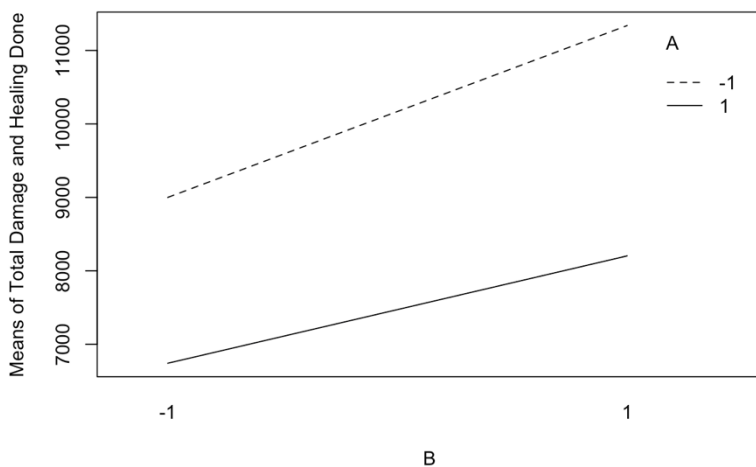


Figure 8: Sensitivity vs Play Style

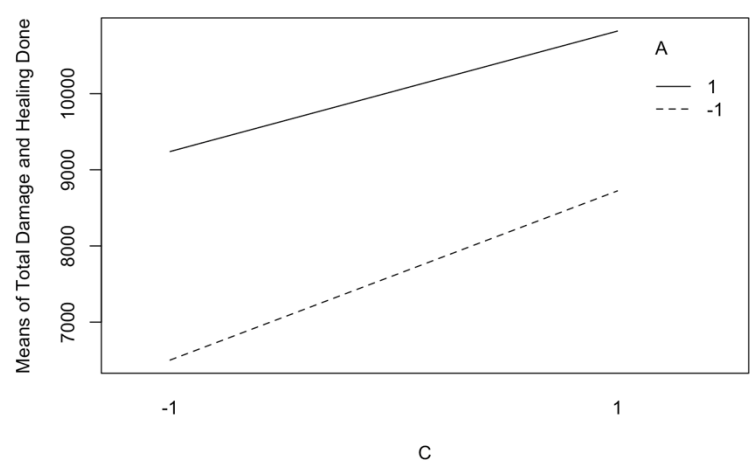


Figure 9: Positioning vs Play Style

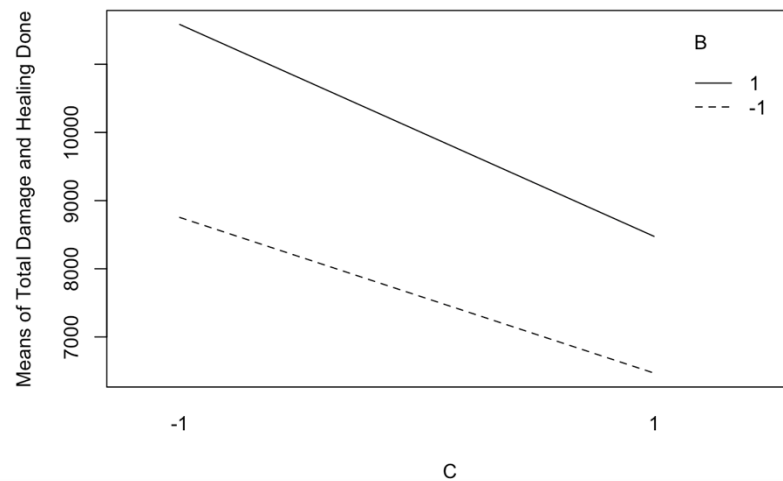


Figure 7 shows us that a lower sensitivity will increase total damage and healing done more than a high sensitivity for both positioning types. Although A- will increase more, the relationship between A and B is still synergistic. Figure 8 also shows us that a lower sensitivity will have a higher increase on total damage done for both play styles. Although A- has the higher increase, the total damage and healing done by A+ remains higher. Figure 9 shows that there is an about equal decrease in total damage and healing done between positioning type and play style. This indicates that it is not recommended to play flank angles when prioritizing both damage and healing.

Conclusion

Through all of our analysis of data, it appears to be clear that the recommended settings for playing *Overwatch 2* with the hero Zenyatta is to have a high sensitivity (80%), play in the back line and to focus healing in order to maximize total damage and healing done, given all assumptions are met. The biggest weakness to this experiment is that all runs were done with a single person. Therefore, this conclusion is only helpful to people of similar skill level and play

style to that of myself and is not universal for all players wishing to maximize their Zenyatta play.

R input/output

All programing submitted on separate .html file